b) Unification and Resolution

Aim:

To implement the Unification and Resolution algorithm using PROLOG.

Procedure:

1. Open Notepad (or the SWI-Prolog editor).
2. Write the facts and rules and save the file with **.pl** extension.
3. In SWI-Prolog, load the program using **File -> Consult** and select the file.
4. Type the query at the **?-** prompt, ending it with a full stop (.). Use **;** to get further answers.
5. Observe and record the output.

Prolog Code:

Exercise 1: Sentence Matcher

```
% Facts
sentence(the_cat_sits).
sentence(the_dog_runs).

% Query
% ?- sentence(X).
```

Output:

X = the_cat_sits ; X = the_dog_runs ; false.

Exercise 2: Compound Term Unification

```
% Facts
likes(john, food(pizza, spicy)).
likes(mary, food(pizza, sweet)).

% Query
% ?- likes(john, food(pizza, T)).
```

Output:

T = spicy.

Exercise 3: Conflict Resolution

```
% Facts
rule1(climate_change).
rule2(climate_change).

% Rule
conflict(X) :- rule1(X), rule2(X).

% Query
% ?- conflict(climate_change).
```

Output:

true.

Result:

Thus the Unification and Resolution algorithm was implemented using PROLOG and executed successfully.